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Chapter 1

Setup and Hypotheses

1.1 Standing hypotheses

Definition 1 (Standing hypotheses)**.** In Lean 4 / Mathlib these are:

```
[Group G] [Fintype G] [Field k] [IsAlgClosed k] [Invertible ((Fintype.card G :
                                    k))].
```

The invertibility hypothesis is equivalent to `NeZero ((Nat.card G : k))` and to $\mathrm{char}(k) \nmid |G|$; the prototypical case is $k = \mathbb{C}$. We write $|G|$ for `Fintype.card G`.

1.2 Mathlib API for representations

Definition 2 (Representation)**.** `Representation k G V`, defined as the abbreviation `G →* (V → [k] V)`, lives in `Mathlib.RepresentationTheory.Basic`. Throughout the blueprint `: Representation k G V` stands for the action of G on a k-module V.

Definition 3 (ρ.asModule)**.** Each `Representation k G V` has a canonically associated `(MonoidAlgebra k G)`-module `.asModule`; this is the same carrier V with $k[G]$-action $\left(\sum_g a_g g\right) \cdot v = \sum_g a_g\, \rho(g) v$. Conversely, any `Module (MonoidAlgebra k G) M` gives a representation `Representation.ofModule M`.

Definition 4 (Subrepresentation)**.** A `Subrepresentation` is a k-submodule of V closed under $\rho(g)$ for every $g \in G$ (file `Mathlib.RepresentationTheory.Subrepresentation`). The map `Subrepresentation.subrepresentationSubmoduleOrderIso` is an order-preserving bijection between `Subrepresentation` and `Submodule (MonoidAlgebra k G) .asModule`, so subrepresentations and $k[G]$-submodules are interchangeable.

Definition 5 (Intertwining map)**.** `.IntertwiningMap` (file `Mathlib.RepresentationTheory.Intertwining`) is the bundled type of k-linear maps $V \to W$ that commute with ρ and σ; it has `FunLike` and `LinearMapClass` instances and a k-module structure. `Representation.IntertwiningMap.equivAlgEnd` identifies `.IntertwiningMap` with the k-algebra of G-equivariant endomorphisms.

Definition 6 (Isomorphism of representations)**.** `.Equiv` is the bundled type of k-linear isomorphisms $V \simeq W$ that intertwine ρ and σ (`Mathlib.RepresentationTheory.Intertwining`).

Definition 7 (Irreducible representation)**.** `.IsIrreducible`, an abbreviation for `IsSimpleOrder (Subrepresentation )`, says ρ is non-trivial and has no proper non-zero subrepresentation (`Mathlib.RepresentationTheory.Irreducible`). `Representation.irreducible_iff_isSimpleModule_asModule` provides the equivalence with `IsSimpleModule (MonoidAlgebra k G) .asModule`, and the corresponding instance is registered (so simplicity of the $k[G]$-module is automatic).

Definition 8 (`FDRep k G`)**.** `FDRep k G` (`Mathlib.RepresentationTheory.FDRep`) is the k-linear abelian symmetric monoidal category of finite-dimensional k-linear representations of G. For V : `FDRep k G` the underlying space is V (via coercion), the action is `V.` , and `[CategoryTheory.Simple V]` encodes simplicity in this category. All the character-theoretic Mathlib lemmas (Section 4.1) are stated in `FDRep`.

Chapter 2

Maschke and Semisimplicity

The whole chapter is essentially the citation `MonoidAlgebra.Submodule.instIsSemisimpleModule`; we just state its consequences explicitly as named statements so subsequent chapters can refer to them.

2.1 Maschke's theorem in Mathlib

Theorem 9 (Maschke: every $k[G]$-module is semisimple)**.** *Under the standing hypotheses (Definition 1), for every `Module (MonoidAlgebra k G) V` we have `IsSemisimpleModule (MonoidAlgebra k G) V`: every $k[G]$-submodule has a $k[G]$-complement.*

Proof. This is the registered instance `MonoidAlgebra.Submodule.instIsSemisimpleModule` in `Mathlib.RepresentationTheory.Maschke`, with hypotheses `[Field k] [Group G] [Finite G] [NeZero (Nat.card G : k)]`. The Mathlib proof uses `LinearMap.equivariantProjection` (averaging π over G) plus `LinearMap.equivariantProjection_condition`. □

Corollary 10 ($k[G]$ is a semisimple ring)**.** *`IsSemisimpleRing (MonoidAlgebra k G)` holds.*

Proof. By definition, `IsSemisimpleRing R := IsSemisimpleModule R R`. Specialise Theorem 9 to $V :=$ `MonoidAlgebra k G`. Formalised as `PeterWeyl.instIsSemisimpleRing_groupAlgebra` via `inferInstance`. □

Corollary 11 (Every V : `FDRep k G` is semisimple)**.** *For every V : `FDRep k G`, `V. .asModule` is a semisimple `MonoidAlgebra k G`-module and is finite as a $k[G]$-module.*

Proof. Direct from Theorem 9; finiteness over $k[G]$ follows from finiteness over k (which is built into `FDRep`) since k is a subring of $k[G]$. □

Chapter 3

Schur's Lemma in Mathlib

Schur is fully formalised in two complementary forms; we just cite both.

3.1 Categorical form (`FDRep`)

Theorem 12 (Schur, finrank form)**.** *Let V, W : `FDRep k G` both carry `[CategoryTheory.Simple]`. Then*

$$\operatorname{finrank}_k(V \to W) \;=\; \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

This is `FDRep.finrank_hom_simple_simple` (file `Mathlib.RepresentationTheory.FDRep`).

Proof. Cited directly. The Mathlib proof is the standard one: a non-zero map between simples has trivial kernel and trivial cokernel hence is an isomorphism, so $V \to W$ is either 0 or $\cong V \to V$, and on $V \to V$ algebraic closedness gives an eigenvalue λ, forcing $T - \lambda \mathrm{id} = 0$. ☐

3.2 Bundled `IntertwiningMap` form

Theorem 13 (Schur, scalar form)**.** *For ρ irreducible and finite-dimensional over an algebraically closed field k, the canonical algebra map*

$$k \;\longrightarrow\; \operatorname{End}_G(\rho) \;=\; \rho.\mathrm{IntertwiningMap}\,\rho$$

is bijective. Hence $\operatorname{End}_G(\rho) \simeq_{k\text{–alg}} k$.

Proof. This is `Representation.IsIrreducible.algebraMap_intertwiningMap_bijective_of_isAlgClosed` (`Mathlib.RepresentationTheory.Irreducible`). ☐

Theorem 14 (Schur, dimension form)**.** *For ρ as above,* $\operatorname{finrank}_k \rho.\mathrm{IntertwiningMap}\,\rho = 1$.

Proof. This is `Representation.IsIrreducible.finrank_intertwiningMap_self`. ☐

Theorem 15 (Schur for distinct irreducibles)**.** *If ρ, σ are both irreducible and there is no G-equivariant isomorphism between them (i.e. `IsEmpty ( .Equiv )`), then `Subsingleton ( .IntertwiningMap )`; equivalently, every intertwiner $\rho \to \sigma$ is zero.*

Proof. This is the registered instance `Representation.IsIrreducible.instSubsingletonIntertwiningMapOfIsEm`
☐

Chapter 4

Characters and Their Orthogonality

The character theory we need (definition, conjugation invariance, isomorphism invariance, the average formula, and Schur orthogonality) is fully formalised in `Mathlib.RepresentationTheory.Character`; we list the relevant declarations.

4.1 Mathlib's character API

Definition 16 (Character of V : `FDRep k G`)**.** $\chi_V(g) = \mathrm{tr}(V.\rho(g))$, defined as `FDRep.character V g`.

Lemma 17 ($\chi_V(1) = \dim V$)**.**

Proof. `FDRep.char_one`. ☐

Lemma 18 (Class function)**.** $\chi_V(hgh^{-1}) = \chi_V(g)$.

Proof. `FDRep.char_conj`. ☐

Lemma 19 (Iso-invariance)**.** $V \cong W$ *in* `FDRep k G` $\Rightarrow \chi_V = \chi_W$.

Proof. `FDRep.char_iso`. ☐

Theorem 20 (Average of χ_V)**.** $|G|^{-1} \sum_{g \in G} \chi_V(g) = \mathrm{finrank}_k V^G$.

Proof. `FDRep.average_char_eq_finrank_invariants`. ☐

Theorem 21 (Hom-space dimension via characters)**.** $|G|^{-1} \sum_g \chi_V(g)\chi_W(g^{-1}) = \mathrm{finrank}_k(V \to W)_G$.

Proof. Apply Theorem 20 to the representation $\mathrm{Hom}_k(V, W)$ with G-action $g \cdot T := V.\rho_W(g) \circ T \circ V.\rho_V(g)^{-1}$. Its character at g equals $\chi_V(g^{-1})\chi_W(g)$, and its G-invariants are exactly $(V \to W)_G$. (In Mathlib this calculation is built into the proof of `FDRep.char_orthonormal` via the Hom-space description as a tensor product of dual and target.) ☐

Theorem 22 (Character orthonormality)**.** *For* V, W : `FDRep k G` *both simple,*

$$|G|^{-1} \sum_{g} \chi_V(g) \chi_W(g^{-1}) \;=\; \begin{cases} 1 & V \cong W, \\ 0 & \textit{otherwise.} \end{cases}$$

Proof. `FDRep.char_orthonormal`. ∎

Chapter 5

Isotypic Decomposition via Mathlib

The full isotypic-component theory is in `Mathlib.RingTheory.SimpleModule.Isotypic`. Specialised to $R :=$ `MonoidAlgebra k G` and the standing hypotheses, every $k[G]$-module is a sum of its isotypic components, each of which is a finite direct sum of copies of one simple module.

5.1 Mathlib's isotypic API

Definition 23 (`isotypicComponent`)**.** For a simple R-module S, `isotypicComponent R M S` is the sum of all submodules of M isomorphic to S (`Mathlib.RingTheory.SimpleModule.Isotypic`).

Theorem 24 (Isotypic components are independent and exhaust)**.** *For any `IsSemisimpleModule R M`, the family `isotypicComponents R M` is `sSupIndep` and its `sSup` is $\top$. Combined with `IsIsotypic.linearEquiv_fun`, every isotypic component is isomorphic to S^n for the corresponding simple S and some $n \in \mathbb{N}$.*

Proof. Cited from Mathlib: `sSupIndep_isotypicComponents` and `sSup_isotypicComponents` (in `Mathlib.RingTheory.SimpleModule.Isotypic`); the per-component description is `IsIsotypic.linearEquiv_fu`
□

Corollary 25 (Module-form Peter–Weyl, Mathlib version)**.** *For any `IsSemisimpleModule R M` with `Module.Finite R M`, there exist $n \in \mathbb{N}$, simple submodules $(S_i)_{i<n}$ of M, and dimensions $(d_i)_{i<n}$ with each $d_i \neq 0$, such that*

$$\mathrm{End}_R M \;\simeq_{R_0\text{-}alg} \prod_{i<n} \mathrm{Mat}_{d_i \times d_i}(\mathrm{End}_R S_i).$$

Proof. This is `IsSemisimpleModule.exists_end_algEquiv_pi_matrix_end` in `Mathlib.RingTheory.SimpleModule`
□

Chapter 6

The Peter–Weyl Theorem for Finite Groups

This is now a one-step application of Wedderburn–Artin to the regular representation, plus an algebraic-closedness step to identify the division algebras with k.

6.1 The regular representation

Definition 26 (Left regular representation)**.** `Representation.leftRegular k G : Representation k G ((G → k))`. Its bundled form is `Rep.ofMulAction k G G : Rep k G`, with the same underlying $k[G]$-module as `MonoidAlgebra k G` acting on itself by left multiplication.

Remark 1 (Character of the regular representation)**.** *In the classical proof of Peter–Weyl one would compute $\chi_{k[G]}(g) = |G| \cdot [g = 1]$ (since left translation by g acts as a permutation matrix on the basis $\{\delta_h\}$ and has $|G|$ fixed points if $g = 1$, otherwise none) and use Schur orthogonality to deduce that every irreducible appears with multiplicity $\dim V_\xi$ in $k[G]$. The proof in this blueprint does* not *use this calculation; we recover the same conclusion as a consequence of the Wedderburn–Artin decomposition (Theorem 29). We mention it only to connect to the classical exposition.*

6.2 Wedderburn form of $k[G]$

Theorem 27 (Wedderburn–Artin for $k[G]$)**.** *Under the standing hypotheses there exist $n \in \mathbb{N}$, division k-algebras $(D_i)_{i<n}$ each finite over k, and $(d_i)_{i<n}$ with each $d_i \neq 0$, together with a k-algebra isomorphism*

$$\mathrm{MonoidAlgebra}\ k\ G \ \simeq_{k\text{-}alg} \prod_{i<n} \mathrm{Mat}_{d_i \times d_i}(D_i).$$

Proof. Apply `IsSemisimpleRing.exists_algEquiv_pi_matrix_divisionRing_finite` with $R_0 := k$, $R :=$ MonoidAlgebra k G. The hypothesis `[Module.Finite k (MonoidAlgebra k G)]` holds because MonoidAlgebra $k\,G \simeq (G \to_0 k)$ has k-dimension $|G| < \infty$. The hypothesis `[IsSemisimpleRing R]` is Corollary 10. ☐

6.3 Algebraic closedness collapses the D_i to k

Lemma 28 (Finite-dim division algebra over alg-closed k equals k)**.** *Let k be an algebraically closed field and D a finite-dimensional division k-algebra. Then the canonical map $k \to D$ is a k-algebra isomorphism.*

Proof. The canonical map $\iota : k \to D$ is injective (it has trivial kernel since k is a field and $\iota(1) = 1 \neq 0$). For surjectivity, take $d \in D$. Since D is finite-dimensional over k, the powers $1, d, d^2, \ldots$ are k-linearly dependent, so d satisfies some non-zero polynomial $f \in k[X]$. Pick f of minimal degree; it is then irreducible in $k[X]$ (else a proper factorisation would give zero-divisors in D, contradicting D a division ring). Since k is algebraically closed, every irreducible polynomial in $k[X]$ has degree 1, so $\deg f = 1$ and $d \in \iota(k)$. The Lean formalisation (`PeterWeyl.divisionAlgebra_algEquiv_of_isAlgClosed`) packages this as `AlgEquiv.ofBijective (Algebra.ofId k D)`, with surjectivity supplied by `IsAlgClosed.algebraMap_bijective_of_isIntegral` (using `Algebra.IsIntegral.of_finite`) and injectivity by `(algebraMap k D).injective`. □

Theorem 29 (Peter–Weyl for finite groups, ring form)**.** *Under the standing hypotheses there exist $n \in \mathbb{N}$ and $(d_i)_{i<n}$ with each $d_i \neq 0$, together with a k-algebra isomorphism*

$$\text{MonoidAlgebra } k\ G \ \simeq_{k\text{-}alg} \prod_{i<n} \text{Mat}_{d_i \times d_i}(k).$$

Proof. Apply Theorem 27 to obtain $\Phi : k[G] \simeq_k \prod_i \text{Mat}_{d_i}(D_i)$ with each D_i a finite-dimensional division k-algebra. Apply Lemma 28 componentwise to obtain $D_i \simeq k$ as k-algebras, and lift through `AlgEquiv.mapMatrix` via `AlgEquiv.piCongrRight` to get $\prod_i \text{Mat}_{d_i}(D_i) \simeq \prod_i \text{Mat}_{d_i}(k)$. Compose. The Lean formalisation (`PeterWeyl.groupAlgebra_algEquiv_pi_matrix`) follows this structure exactly. □

6.4 Sum-of-squares dimension formula

Corollary 30 (Sum of squares of dimensions)**.** *Under the standing hypotheses, with $(d_i)_i$ the dimensions appearing in Theorem 29,*

$$|G| \ = \sum_{i<n} d_i^2.$$

Proof. Take $\dim_k$ of both sides of the isomorphism in Theorem 29: the left side is $\dim_k k[G] = |G|$ (`Module.finrank_finsupp_self`); the right side, by `Module.finrank_pi_fintype` and `Module.finrank_matrix`, is $\sum_i \dim_k \text{Mat}_{d_i}(k) = \sum_i d_i^2$. □

6.5 Module-form Peter–Weyl

Theorem 31 (Module form of Peter–Weyl)**.** *Under the standing hypotheses there exist $n \in \mathbb{N}$, simple subrepresentations $(V_i)_{i<n}$ of the regular representation, and multiplicities $(d_i)_{i<n}$ with each $d_i = \dim_k V_i \neq 0$, such that*

$$\text{MonoidAlgebra } k\ G \ \simeq \ \bigoplus_{i<n} V_i^{\oplus d_i}$$

as $k[G]$-modules. Equivalently: every irreducible appears in $k[G]$ with multiplicity equal to its k-dimension, and these are all the irreducibles.

Proof. Apply Corollary 25 to $R := \mathrm{MonoidAlgebra}\ k\ G$, $R_0 := k$, and $M := \mathrm{MonoidAlgebra}\ k\ G$ (a finite $k[G]$-module since $|G| < \infty$ and so a finite k-module). This produces simple subrepresentations V_i and dimensions d_i together with a k-algebra isomorphism $\mathrm{End}_{k[G]}\, k[G] \simeq \prod_i \mathrm{Mat}_{d_i}(\mathrm{End}_{k[G]} V_i)$. Now $\mathrm{End}_{k[G]}\, k[G] \simeq k[G]^{\mathrm{op}}$ canonically ($\texttt{moduleEndSelf} = \texttt{MulOpposite.opAlgEquiv}$), and by Schur (Theorem 13) $\mathrm{End}_{k[G]} V_i \simeq k$. The isomorphism then becomes $k[G]^{\mathrm{op}} \simeq \prod_i \mathrm{Mat}_{d_i}(k)$, recovering Theorem 29 (and its module form by reading off the decomposition of the standard $k[G]$-module on the right). The dimension identification $d_i = \dim_k V_i$ comes from comparing dimensions of $V_i^{\oplus d_i}$ and $\mathrm{Mat}_{d_i}(k)$ via the equivalence $\mathrm{Mat}_{d_i}(k) \cong V_i^{\oplus d_i}$ as left $\mathrm{Mat}_{d_i}(k)$-modules. $\square$

Chapter 7

Character-Theoretic Consequences

These follow immediately from the Mathlib character API combined with the isotypic decomposition.

7.1 Multiplicity formula and irreducibility test

Theorem 32 (Multiplicity formula)**.** *Let V : `FDRep k G` and ξ a simple V_ξ : `FDRep k G`. Write m_ξ for the multiplicity of V_ξ in the isotypic decomposition of V (Theorem 24). Then*

$$m_\xi \;=\; |G|^{-1}\sum_g \chi_V(g)\chi_{V_\xi}(g^{-1}) \;=\; \mathrm{finrank}_k(V_\xi \to V)_G.$$

Proof. Decompose $V \cong \bigoplus_\eta V_\eta^{\oplus m_\eta}$ via Theorem 24. Then $\mathrm{finrank}_k(V_\xi \to V)_G = \sum_\eta m_\eta \mathrm{finrank}_k(V_\xi \to V_\eta)_G = m_\xi$ by Theorem 12. The first equality is then Theorem 21. □

Corollary 33 (Irreducibility test)**.** *V : `FDRep k G` is simple iff $|G|^{-1}\sum_g \chi_V(g)\chi_V(g^{-1}) = 1$.*

Proof. Decompose $V \cong \bigoplus_\xi V_\xi^{\oplus m_\xi}$ and expand. By Theorem 22, $|G|^{-1}\sum_g \chi_V(g)\chi_V(g^{-1}) = \sum_\xi m_\xi^2$. This integer is 1 iff exactly one m_ξ is 1 and the rest are 0, iff $V \cong V_\xi$ is simple. □

Corollary 34 (Characters separate isomorphism classes)**.** *$V \cong W$ in `FDRep k G` iff $\chi_V = \chi_W$.*

Proof. $\Rightarrow$: Lemma 19. $\Leftarrow$: by Theorem 32 the multiplicities $m_\xi(V)$ and $m_\xi(W)$ depend only on the characters, so they coincide; uniqueness of the isotypic decomposition (Theorem 24) then forces $V \cong W$. □

7.2 Class functions and the second orthogonality

Definition 35 (Class functions)**.** $\mathrm{Cl}(G) := \{f : G \to k \mid \forall g, h.\ f(hgh^{-1}) = f(g)\}$, a k-subspace of k^G.

Lemma 36 ($\dim \mathrm{Cl}(G) = \#$conjugacy classes)**.** $\dim_k \mathrm{Cl}(G) = |\mathrm{ConjClasses}\ G|$.

Proof. The indicator functions of conjugacy classes form a basis of $\mathrm{Cl}(G)$. $\square$

Theorem 37 (Irreducible characters span $\mathrm{Cl}(G)$)**.** *The set* $\{\chi_V : V \in$ `FDRep k G`, `Simple` $V\}$ *spans* $\mathrm{Cl}(G)$ *over* k*.*

Proof. Under Theorem 29 the centre of $k[G]$ pulls back to the centre of $\prod_i \mathrm{Mat}_{d_i}(k)$, which is $\prod_i k$, of k-dimension n (the number of simples). On the other hand the centre of $k[G]$ identifies with $\mathrm{Cl}(G)$ via $\sum_g a_g g \leftrightarrow (g \mapsto a_g)$ (an element commutes with every $h \in G$ iff its support function is conjugation-invariant). Hence $\dim_k \mathrm{Cl}(G) = n$, the number of simples. Since the irreducible characters are orthonormal (Theorem 22), there are n of them, and they live in $\mathrm{Cl}(G)$ by Lemma 18; hence they span (a linearly-independent set of size equal to the dimension is a basis). $\square$

Theorem 38 (Number of irreducibles = number of conjugacy classes)**.** *The number of isomorphism classes of simple objects of* `FDRep k G` *equals* $|\mathrm{ConjClasses}\ G|$*.*

Proof. By Theorem 37 and Theorem 22, the irreducible characters form a basis of $\mathrm{Cl}(G)$; their number is therefore $\dim_k \mathrm{Cl}(G) = |\mathrm{ConjClasses}\ G|$ (Lemma 36). $\square$

Chapter 8

Mathlib-Status Summary

Already in Mathlib (`[ML]`, with `\leanok`). The following declarations are direct citations: `Representation`, `Representation.asModule`, `Subrepresentation` (and its order isomorphism with $k[G]$-submodules), `Representation.IntertwiningMap`, `Representation.Equiv`, `Representation.IsIrreduci` (and the equivalence with `IsSimpleModule`), `FDRep`, `Representation.leftRegular`, `Rep.ofMulAction`, `MonoidAlgebra.Submodule.instIsSemisimpleModule` (Maschke), `IsSemisimpleRing` as a derived instance, `FDRep.finrank_hom_simple_simple` (Schur), `Representation.IsIrreducible.algebraMap_intert` (scalar Schur), `Representation.IsIrreducible.finrank_intertwiningMap_self` (dimension Schur), `Representation.IsIrreducible.instSubsingletonIntertwiningMapOfIsEmptyEquiv` (Schur for distinct irreducibles), `FDRep.character` and the basic identities `FDRep.char_one`, `FDRep.char_conj`, `FDRep.char_iso`, `FDRep.average_char_eq_finrank_invariants`, `FDRep.char_orthonormal`, `isotypicComponent`, `IsIsotypic`, `sSupIndep_isotypicComponents`, `sSup_isotypicComponents`, `IsIsotypic.linearEquiv_fun`, `IsSemisimpleModule.exists_end_algEquiv_pi_matrix_end`, `IsSemisimpleRing.exists_algEquiv_pi_matrix_divisionRing_finite`.

Trivial wrappers (`[~ML]`). `FDRep.isSemisimpleModule_asModule` (Corollary 11), `Representation.multiplicit` (Theorem 32), `FDRep.irreducible_iff_inner_self_eq_one` (Corollary 33), `FDRep.iso_iff_character_eq` (Corollary 34), `ClassFunction` and `ClassFunction.finrank_eq_conjClasses`.

Genuinely new contributions (`[NEW]`). There are five:

1. `Representation.finrank_hom_eq_inner_char` — the hom-space dimension formula $|G|^{-1}\sum_g \chi_V(g)\chi_W(g^{-1}) =$ $\mathrm{finrank}_k(V \to W)_G$ (Theorem 21). Currently appears in Mathlib only as an intermediate step inside the proof of `FDRep.char_orthonormal`; extract as a standalone lemma.

2. `PeterWeyl.divisionAlgebra_algEquiv_of_isAlgClosed` — every finite-dimensional division algebra over an algebraically closed field equals the field (Lemma 28). The Lean proof bundles `IsAlgClosed.algebraMap_bijective_of_isIntegral` (for surjectivity) with `(algebraMap k D).injective` via `AlgEquiv.ofBijective`.

3. `PeterWeyl.groupAlgebra_algEquiv_pi_matrix` (Theorem 29). Combines the Mathlib Wedderburn–Artin output with the previous lemma; one line of `simp` and apply.

4. `PeterWeyl.sum_sq_dim_eq_card` (Corollary 30). `finrank` of both sides.

5. `FDRep.span_irreducibleCharacters_eq_top` and the corollary `FDRep.num_simple_eq_num_conjClasses` (Theorems 37, 38). The argument identifies the centre of $k[G]$ with $\mathrm{Cl}(G)$ and reads off dimension from Theorem 29.

What this means. The whole project is essentially a five-page Lean file: `Maschke + WedderburnArtin + algClosed division-algebra collapse`. The proof of Peter–Weyl in this blueprint sidesteps the classical isotypic calculation entirely (which is itself already in Mathlib) and instead routes through the ring-theoretic Wedderburn–Artin theorem.